Processing and Visualising Large DEMs on a Laptop

Progress Report, Phase 4: Terrain Derivatives on Demand from the Pyramid

Team GeoGuessers | Spatial Informatics

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Abstract

A viewer showing a large area cannot use the full-resolution DEM: it reads a coarser pyramid level and computes slope there. No study has measured how wrong such a coarse-level slope is, or how much time it saves (research gap **G2**). We measured both on real Himalayan data: the Chamoli 2 m DEM with pyramid levels from 4 m to 128 m, and an 8 m DEM with levels from 16 m to 256 m. Slope and aspect computed on demand from each level were compared with the full-resolution result averaged over exactly the same ground, on 61 million down to 29 thousand fully valid cells per level. On demand, one 512×512 display tile costs 5–8 ms at every zoom level, 8–450× less than computing it exactly from full resolution, and needs no precomputed storage. The price is a systematic underestimate of slope: on the 2 m data the RMSE grows from 2.3° one level up to 6.4° four levels up and 10.0° six levels up, worst on the steepest ground, where a coarse level misses a quarter or more of the terrain steeper than 45°. Aspect survives much better (median error 2–8°). A pyramid level of the 2 m data is closer to the truth than an independent DEM of the same resolution (HMA 8 m, Copernicus 30 m and 90 m). A precomputed slope pyramid is exact but costs a full pass and 1.0 GB per derivative.

1 Objective

The project studies how to process and visualise very large DEMs on one laptop by reading only what is in view: a pyramid (multi-resolution) representation, derivatives computed on demand, and CPU/GPU processing chosen by the measured bottleneck. Phases 2 and 3 located the bottleneck of full-resolution processing (gap **G1**). This phase addresses gap **G2**: *no study computes terrain derivatives on demand from pyramid levels, or measures the error of coarse-level slope against the time it saves.* The questions are:

1. How far is slope (and aspect) computed from pyramid level k from what full-resolution data would show for the same screen cell, and on which terrain is the error largest?
2. How much time does computing on demand save, compared with computing at full resolution, or with a derivative pyramid built in advance?

2 Data and pyramid

Accuracy needs real data, never upsampled grids, so two real DEMs of the study area are used (Table 1). Each is stored as a Cloud-Optimized GeoTIFF whose overviews, built by GDAL with averaging, form the pyramid; these overviews are exactly what a viewer reads.

Table 1: Data and pyramid levels used.

DEM	Cells	Valid	Pyramid levels (cell size)
Chamoli 2 m composite [1] (main result)	264 M	94%	4, 8, 16, 32, 64, 128 m (levels 1–6)
HMA 8 m mosaic [2], tile-571 (whole area)	169 M	84%	16, 32, 64, 128, 256 m (levels 1–5)

GDAL halves each level's size and rounds down, and every level keeps the full extent. When the grid size is not divisible by 2^k, a level's cell is slightly larger than 2^k fine cells (on Chamoli, level 3 cells cover 8.002 fine rows instead of 8). Our analysis therefore takes each coarse cell's footprint from its geometry (the fine cells whose centres fall inside it), and repeats every measurement on an exact power-of-two pyramid built by ourselves.

3 Method

Error. For every pyramid level k:

- **On demand:** slope (Horn [4]) and aspect computed by our engine's kernel on the level-k DEM, with that level's cell size.
- **Truth:** slope and aspect computed at full resolution, then averaged over exactly the fine cells each coarse cell covers (slope: mean; aspect: circular mean of the fine directions). This is what a screen pixel of that size *should* show: the average steepness of the ground under it.
- **Scored cells:** only cells whose footprint and 8 neighbours contain no nodata at full resolution, so data gaps never enter the comparison. Aspect is scored where the true slope is at least 5° and the fine directions agree (mean resultant length ≥ 0.5; 87–99% of the scored slope cells).
- **Metrics:** RMSE, bias (mean of on-demand minus truth), share of cells off by more than 5° and 10°, all split by true-slope class; share of cells above 30° and 45°.
- **Independent DEMs:** over the Chamoli footprint, HMA 8 m and Copernicus 30 m and 90 m [3] against the same truth, next to the 2 m data averaged to the same resolution.

Time. Three ways to serve a view at level k, measured in one long-running process (as a tile server would be), on the 12 performance-core threads:

- **On demand:** read the view from DEM level k (plus a one-cell halo) and compute slope.
- **Full resolution:** read the same ground at full resolution, compute slope, average to level k: the exact answer without any precomputation.
- **Precomputed:** read the view from a slope pyramid built in advance.

Views are one 512×512-cell display tile at 8 random places per level (each at least 90% valid) and the whole grid at each level. *Cold*: the Windows file cache is emptied before each measurement; *warm*: the file is already cached (median of 3). A fresh dataset handle is opened for every measurement, so GDAL's own block cache never hides the decoding. Medians are reported; the spread (median and IQR) of every timing is kept.

4 Results

4.1 Slope error per pyramid level

- **Slope from a coarse level is too low, and more so the coarser the level.** Averaging the DEM smooths the terrain before slope is computed, which removes steepness; the mean slope shown drops from 99% to 81% of the true mean on Chamoli. This agrees with the long-known decrease of computed slope with grid size [5]; what is new here is the measurement on pyramid levels against the time saved.
- **What matters is how many levels up, not the cell size in metres.** The two DEMs give similar errors for the same number of levels up (within 0.5° up to four levels): about 2.5° RMSE one level up, then on average about 1.5° more per further level.

Table 2: Slope computed on demand from each pyramid level vs the full-resolution truth. Last column: the same with an exact power-of-two pyramid instead of GDAL's overviews.

DEM	Level	Cells scored	RMSE	Bias	>5°	>10°	Shown / true mean	RMSE exact
Chamoli 2 m	4 m	61.4 M	**2.3°**	−0.5°	4.1%	0.7%	0.99	2.3°
	8 m	15.2 M	**4.4°**	−1.2°	15.5%	4.1%	0.97	4.4°
	16 m	3.7 M	**5.6°**	−2.2°	23.4%	7.2%	0.94	5.6°
	32 m	0.91 M	**6.4°**	−3.3°	30.5%	10.3%	0.91	6.4°
	64 m	0.22 M	**7.9°**	−5.0°	43.1%	17.1%	0.87	7.8°
	128 m	51 k	**10.0°**	−7.3°	57.8%	28.2%	0.81	9.9°
HMA 8 m	16 m	29.6 M	**2.6°**	−0.7°	6.6%	0.9%	0.98	2.4°
	32 m	6.5 M	**4.0°**	−1.6°	15.6%	3.0%	0.95	3.9°
	64 m	1.3 M	**5.3°**	−2.9°	26.3%	6.9%	0.90	5.3°
	128 m	0.23 M	**6.9°**	−4.5°	40.5%	13.3%	0.83	6.9°
	256 m	29 k	**9.0°**	−6.8°	58.7%	24.9%	0.73	8.8°

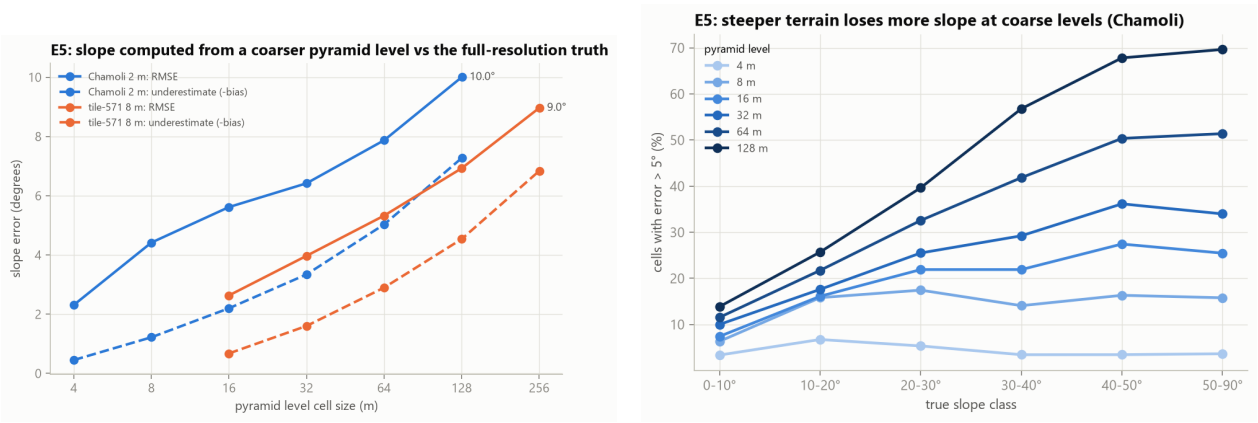


Figure 1: Left: slope RMSE and underestimate (−bias) per pyramid level for both DEMs. Right: share of cells off by more than 5° by true-slope class (Chamoli): steep terrain loses the most.

- **GDAL's overviews behave like an exact pyramid**: within 0.2° RMSE at every level, although their cells are not exactly 2^k fine cells. A tile server can use COG overviews as they are.

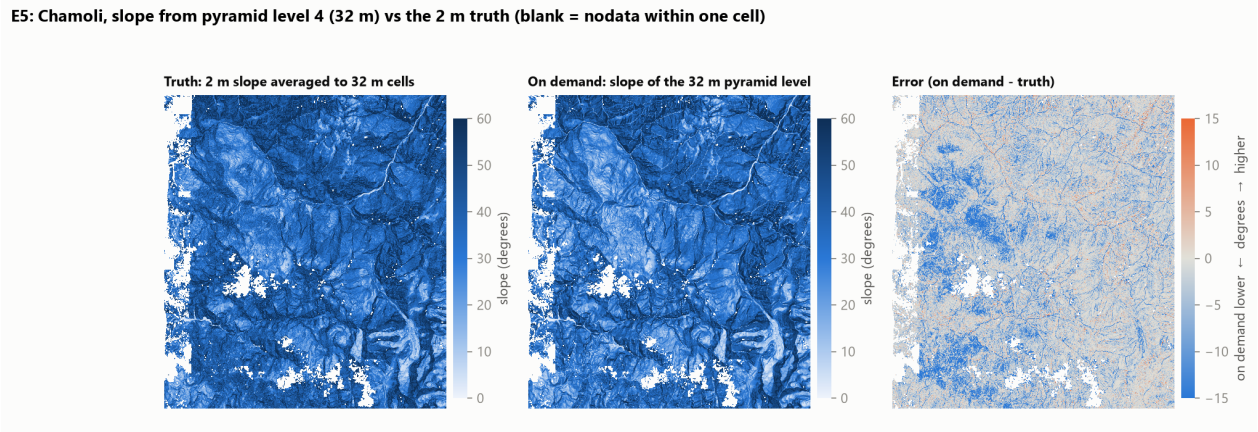


Figure 2: Chamoli at 32 m. Left: truth (2 m slope averaged to 32 m cells). Middle: slope computed on demand from the 32 m pyramid level. Right: error; blue = on demand lower. The on-demand map keeps every valley, ridge and landslide scar but is lower almost everywhere, most on rough, broken slopes.

4.2 Steep terrain and slope thresholds

Table 3: Chamoli: bias by true-slope class, and the share of cells at or above 45° in truth and on demand.

Level	0–10°	10–20°	20–30°	30–40°	40–50°	50–90°	≥45° truth	≥45° shown
4 m	−0.1	−0.2	−0.3	−0.3	−0.6	−0.8	33.2%	32.1%
8 m	−0.0	−0.2	−0.7	−1.0	−1.6	−2.0	32.3%	29.2%
16 m	−0.5	−0.8	−1.4	−2.0	−2.8	−3.0	30.8%	25.7%
32 m	−0.9	−1.5	−2.6	−3.3	−4.0	−3.9	29.3%	21.9%
64 m	−0.7	−1.9	−3.5	−5.1	−6.0	−5.9	28.4%	16.5%
128 m	−0.8	−2.4	−4.3	−7.3	−8.8	−8.9	27.4%	**10.1%**

Steep cells lose the most: at 32 m the bias is −0.9° on 0–10° ground but −4.0° on 40–50° ground, and 36% of the 40–50° cells are off by more than 5° (10% on flat ground). Many uses of slope are a threshold (landslide susceptibility, "too steep to build"). One level up, a threshold map at 30° or 45° still agrees on 97% of cells; four levels up a quarter of the terrain steeper than 45° has disappeared, and six levels up almost two thirds.

4.3 Aspect

The direction a slope faces is set by large landforms and survives smoothing. On Chamoli the median aspect error is 1.8° at 4 m, 3.8° at 8 m, 6.3° at 32 m and 8.5° at 128 m; only 1.4% (4 m) to 13.7% (128 m) of cells are off by more than 22.5°, the width of one 16-point compass sector. The 8 m DEM gives the same picture (median 2.8° at 16 m to 9.9° at 256 m).

4.4 Pyramid error vs differences between data sources

At every resolution the pyramid of the 2 m data is closer to the truth than a DEM measured natively at that resolution by another sensor: 4.5° vs 8.6° for HMA 8 m, 6.3° vs 9.4° for Copernicus 30 m, 8.7° vs 9.9° for Copernicus 90 m (Figure 3). At 8 m the difference between the two sources (7.3° RMSE) is larger than what the pyramid loses; at 90 m the sources agree well (3.3°) and most of the

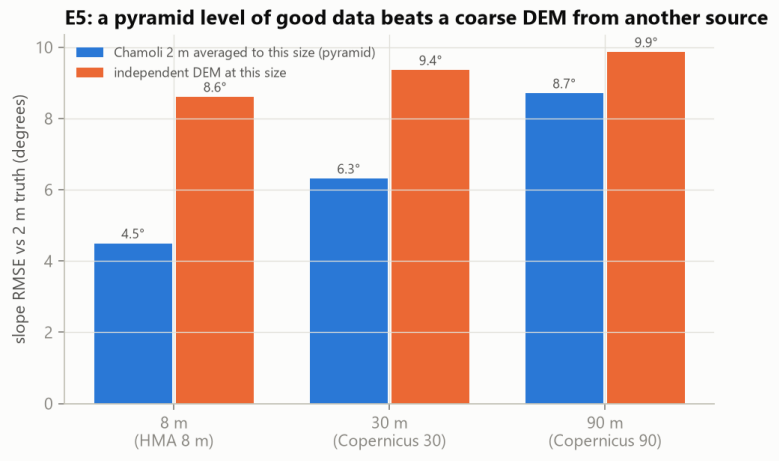


Figure 3: Slope RMSE against the 2 m truth over the Chamoli footprint: the 2 m data averaged to 8, 30 and 90 m (a pyramid level) vs an independent DEM at the same resolution.

error of both is simply the coarse resolution. When a high-resolution DEM exists, its pyramid is the better coarse product.

4.5 Time per view

Table 4: One 512 × 512 display tile, median of 8 places (ms, warm / cold), and the time saved by computing on demand instead of from full resolution (warm).

DEM	Level	On demand	Full resolution + average	Precomputed	Time saved
Chamoli	2 m (level 0)	5.2 / 6.7	= on demand	4.1 / 5.2	–
	4 m	7.2 / 7.6	56 / 55	5.0 / 6.1	**7.8**×
	8 m	6.4 / 7.3	192 / 194	4.3 / 5.3	**30**×
	16 m	6.6 / 7.3	624 / 626	3.8 / 4.9	**94**×
	32 m	6.4 / 6.8	2867 / 2823	3.8 / 4.8	**447**×
HMA 8 m	16 m	6.9 / 7.1	52 / 57	4.2 / 5.3	7.6×
	32 m	7.2 / 7.5	186 / 192	4.5 / 6.2	26×
	64 m	7.1 / 7.6	380 / 400	4.1 / 5.3	54×
	128 m	6.2 / 7.0	1186 / 1204	3.2 / 4.1	193×

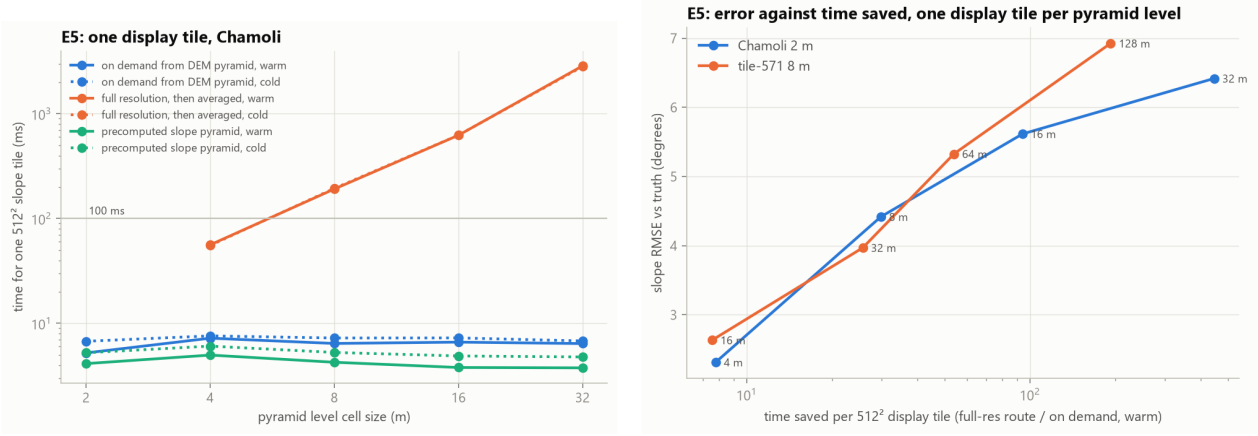


Figure 4: Left: time for one display tile per pyramid level (Chamoli), three routes, cold and warm. Right: the G2 trade-off: slope RMSE against the time saved per tile, one point per pyramid level.

- **On demand costs the same at every zoom level** (5–8 ms per tile): a tile always means reading 514^2 cells and running the kernel on them. The exact full-resolution route reads 4× more ground per level and grows 2–5× per level, to 2.9 s for one tile at 32 m. For the whole Chamoli area at 32 m, on demand takes 13 ms against 4.7 s.
- **Cold vs warm hardly matters** on this NVMe SSD: at most 1.5 ms per tile for on demand.
- **Precomputed is about 1–3 ms faster per tile** (it skips the slope kernel, about 0.5 ms, and the window preparation). On demand and precomputed both stay far below the ∼100 ms at which panning and zooming start to feel slow; the full-resolution route exceeds it from two levels up.

The cost of precomputing. A slope pyramid built in advance is exact and slightly faster per tile, but it costs a full pass for every derivative: for Chamoli 10.6 s (1.6 s slope, 9.0 s compression and overviews) and 1.0 GB, more than the DEM pyramid itself (650 MB), because slope compresses worse than elevation. The ten `gdaldem` configurations would need about 10 GB, rebuilt whenever a parameter (sun angle, z-factor, method) or the DEM changes. On demand needs none of it.

5 Discussion

The G2 trade-off. Each pyramid level up saves 2–5× in time per view (8–450× per tile for one to four levels on Chamoli, 8–190× on the 8 m DEM) and costs about 2.5° slope RMSE for the first level and about 1.5° more per further level on average, as a systematic underestimate that is largest on steep ground. Aspect costs much less (median 2–8°).

Implications for the interactive viewer (Phase 5).

- Choose the pyramid level so that one level cell is about one screen pixel: the on-demand value is then close to what that pixel should show, and every tile costs 5–8 ms, of which about 0.5 ms is the slope kernel.
- Precompute only the one or two most used products (slope), if storage allows; compute everything else on demand, including any hillshade sun angle.
- Add a decoded-tile cache: re-reading compressed blocks, not the kernel, is the main cost of a tile.
- For analysis rather than display, never threshold slope computed from a coarse level: compute the view at full resolution (0.05–2.9 s per tile here) or read a precomputed slope pyramid.

Limitations. One region of steep terrain (mean true slope 38° on Chamoli, 31° on the 8 m area); flatter regions would show smaller errors, and the per-class tables carry over. The truth is itself a DEM with its own noise, and the 8 m DEM's truth is only 8 m. The fully-valid rule scores fewer cells at coarse levels (17% of cells at 256 m on the 8 m DEM, whose data gaps lie in steep terrain). Timing is on one machine with a local NVMe SSD; a network-hosted COG would add latency per request and widen the gap between the routes. The underestimate is systematic, so a per-level correction factor could remove most of the bias, but not the spread; this was not tested.

6 Next step

Interactive viewer (E6): a tile server that computes slope, aspect and hillshade tiles on the fly from the pyramid level matching the zoom, with a decoded-tile cache, serving QGIS and a Cesium globe; tile latency for precomputed vs on-the-fly CPU vs GPU.

References

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[4] Horn, B. K. P. (1981). Hill shading and the reflectance map. *Proceedings of the IEEE*, 69(1), 14–47.

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